

PSTAT 126 Final Project

Julianna Nguyen

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PSTAT 126 Final Project

Part 1: Data Description and Descriptive Statistics

Question 1: Select a random sample

```
prices <- read_csv("Diamonds Prices2022.csv")

## New names:
## Rows: 53943 Columns: 11
## -- Column specification
## ----- Delimiter: "," chr
## (3): cut, color, clarity dbl (8): ...1, carat, depth, table, price, x, y, z
## i Use 'spec()' to retrieve the full column specification for this data. i
## Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## * ' -> '...1'

set.seed(126)
diamonds_sample <- prices %>% sample_n(1000)
```

Question 2: Describe all variables

```
# Structure
str(diamonds_sample)

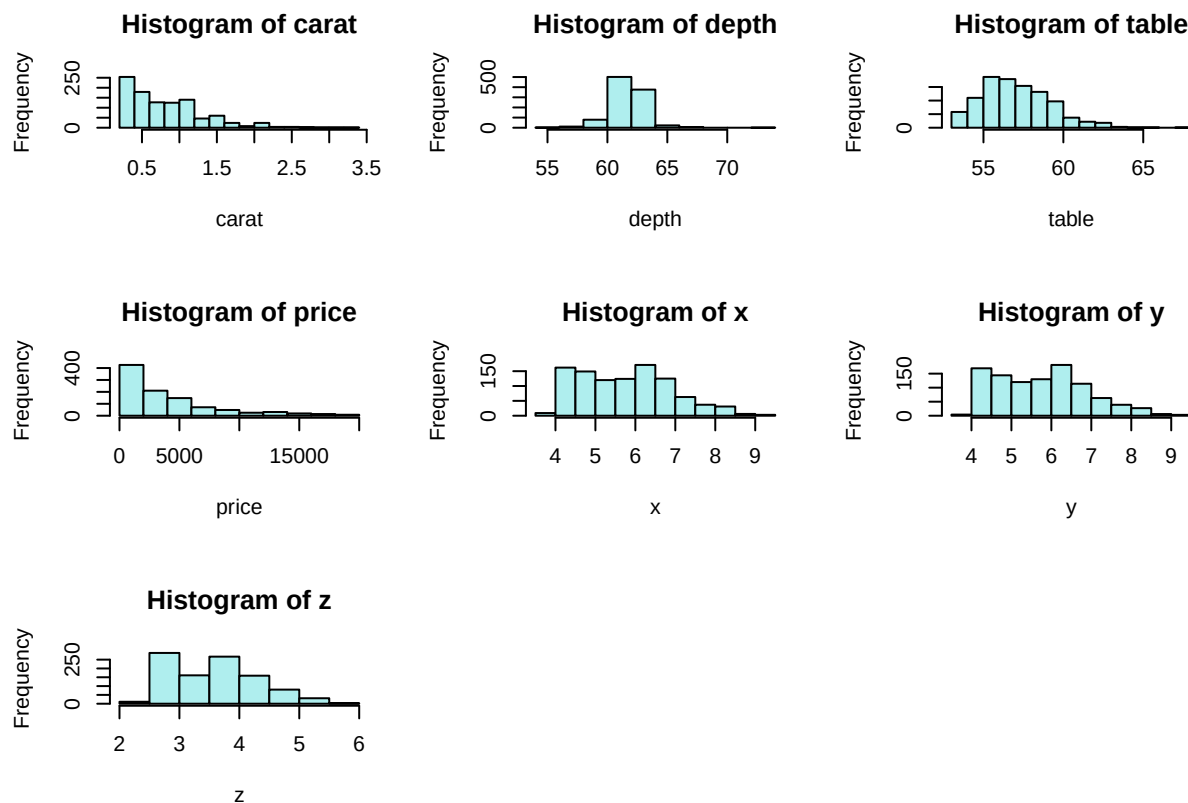
## spc_tbl_ [1,000 x 11] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ ...1 : num [1:1000] 29930 44891 9834 22832 38283 ...
## $ carat : num [1:1000] 0.5 0.5 1.01 3 0.28 0.79 0.72 0.4 0.3 1.12 ...
## $ cut : chr [1:1000] "Fair" "Ideal" "Premium" "Good" ...
## $ color : chr [1:1000] "F" "E" "G" "I" ...
## $ clarity: chr [1:1000] "I1" "VS2" "SI1" "I1" ...
## $ depth : num [1:1000] 63.8 62.1 59.6 57 61.7 61.5 61.7 64 63 63.7 ...
## $ table : num [1:1000] 58 57 61 64 58 55 55 55 55 61 ...
## $ price : num [1:1000] 713 1629 4672 10863 487 ...
## $ x : num [1:1000] 5.08 5.1 6.52 9.38 4.17 5.98 5.76 4.67 4.29 6.54 ...
## $ y : num [1:1000] 4.97 5.05 6.47 9.31 4.19 5.95 5.81 4.7 4.31 6.5 ...
```

```
## $ z      : num [1:1000] 3.21 3.15 3.87 5.33 2.58 3.67 3.57 3 2.71 4.15 ...
## - attr(*, "spec")=
## .. cols(
## ..   ...1 = col_double(),
## ..   carat = col_double(),
## ..   cut = col_character(),
## ..   color = col_character(),
## ..   clarity = col_character(),
## ..   depth = col_double(),
## ..   table = col_double(),
## ..   price = col_double(),
## ..   x = col_double(),
## ..   y = col_double(),
## ..   z = col_double()
## .. )
## - attr(*, "problems")=<externalptr>
```

```
# Summary
summary(diamonds_sample)
```

```
##      ...1      carat      cut      color
## Min.   : 128   Min.   :0.220   Length:1000   Length:1000
## 1st Qu.:12162 1st Qu.:0.400   Class :character   Class :character
## Median :25485 Median :0.710   Mode  :character   Mode  :character
## Mean   :26095 Mean   :0.817
## 3rd Qu.:39800 3rd Qu.:1.040
## Max.   :53923 Max.   :3.240
## clarity      depth      table      price
## Length:1000   Min.    :55.80   Min.    :53.00   Min.    : 379
## Class :character 1st Qu.:61.10   1st Qu.:56.00   1st Qu.: 975
## Mode  :character Median :61.80   Median :57.00   Median : 2596
##                Mean  :61.75   Mean  :57.52   Mean  : 4017
##                3rd Qu.:62.52   3rd Qu.:59.00   3rd Qu.: 5367
##                Max.   :72.20   Max.   :68.00   Max.   :18781
##      x      y      z
## Min.   :3.900   Min.   :3.850   Min.   :2.380
## 1st Qu.:4.737   1st Qu.:4.750   1st Qu.:2.940
## Median :5.740   Median :5.750   Median :3.560
## Mean   :5.774   Mean   :5.777   Mean   :3.566
## 3rd Qu.:6.530   3rd Qu.:6.510   3rd Qu.:4.030
## Max.   :9.440   Max.   :9.400   Max.   :5.850
```

```
# Histogram for Continuous Variables
num_vars <- c("carat", "depth", "table", "price", "x", "y", "z")
par(mfrow = c(3, 3)) # 3x3 layout
for (var in num_vars) {
  hist(diamonds_sample[[var]],
       main = paste("Histogram of", var),
       xlab = var,
       col = "paleturquoise")
}
par(mfrow = c(1, 1))
```



- carat, depth, table, price, x, y, and z are numeric
- cut, color, and clarity are categorical factors

Interpretation

- carat: Right-skewed. Most diamonds in the dataset are relatively small in size. The fewer large diamonds, and those large values form a long tail to the right.
- depth: The depth variable is slightly left-skewed, centered around 60%-62%. It has a symmetric bell shaped with a bit of skewness, indicating consistent proportions in diamond cuts.
- table: Distribution is mildly right-skewed, with a concentration of values between 55 and 60.
- price: The price distribution is strongly right-skewed. Most diamonds are priced under \$5000, but a small number of expensive diamonds create a long tail extending towards a higher price range.
- x, y, z: These are the physical dimensions of diamonds.
 - All three are right-skewed
 - Most diamonds have relatively small dimensions (around 4-6 mm), while a few large diamonds stretch the distribution to the right
 - These variables are likely highly correlated

```
#Bar Plots for Categorical Variables
ggplot(diamonds_sample, aes(x = cut)) +
  geom_bar(fill = "lightblue") +
```

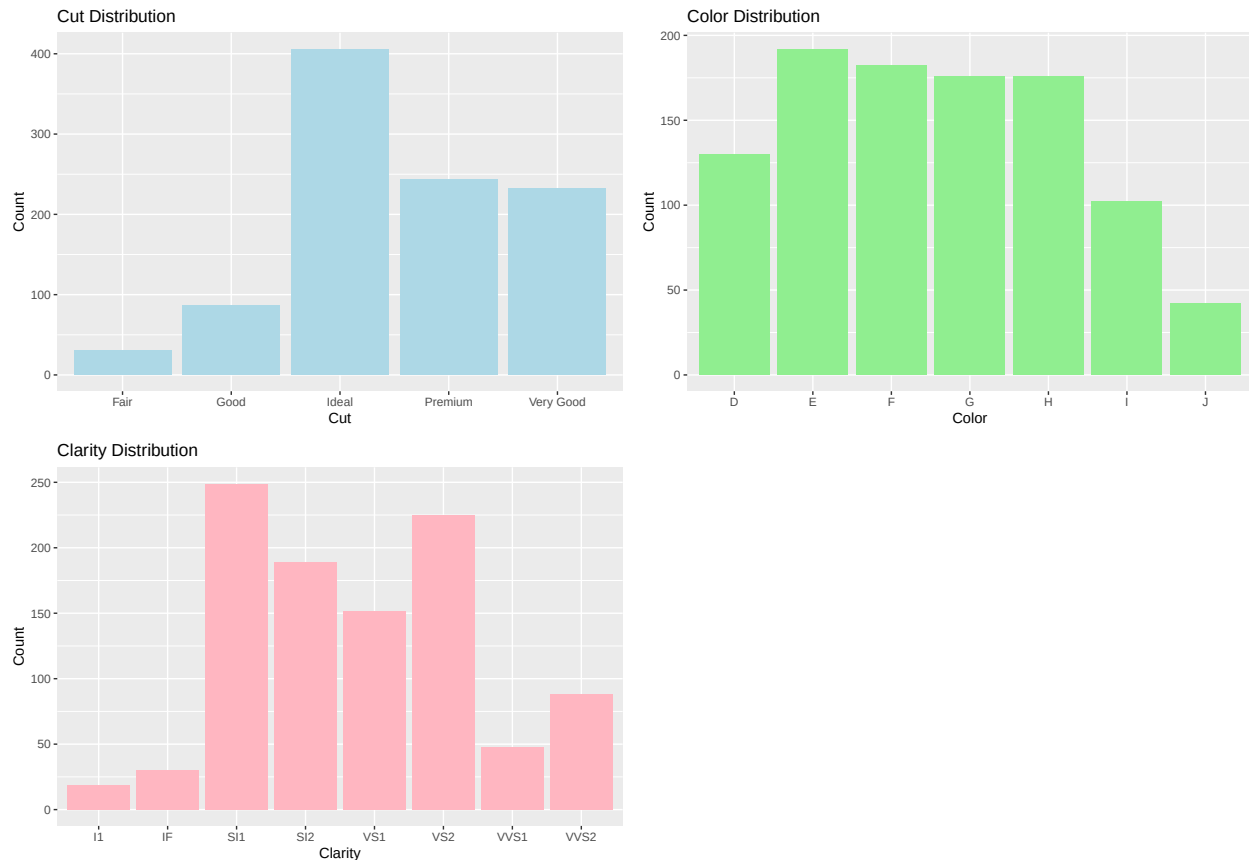
```

labs(title = "Cut Distribution", x = "Cut", y = "Count")

ggplot(diamonds_sample, aes(x = color)) +
  geom_bar(fill = "lightgreen") +
  labs(title = "Color Distribution", x = "Color", y = "Count")

ggplot(diamonds_sample, aes(x = clarity)) +
  geom_bar(fill = "lightpink") +
  labs(title = "Clarity Distribution", x = "Clarity", y = "Count")

```



- cut: The most common cut in the sample is “Ideal,” followed by “Premium,” and “Very Good”
 - “Fair” and “Good” cuts are much less frequent
 - This suggests that most diamonds in this dataset are relatively high cut quality, with lower-quality cuts being rare
- color: Color grades E, F, G, and H dominate the sample
 - D(the best grade) and J(the worst) are less common
 - This reflects that most diamonds are mid to high range color quality, with very high or very low grades being rare in the sample
- clarity: The most common clarity grades are SI1, VS2, and SI2
 - Very high clarity grades like IF and VVS1 are relatively rare
 - I1 is also rare due to lower quality

- The distribution suggests most diamonds are of moderate clarity, with extreme clarity grades less represented

Question 3: Correlation

```
# Select only continuous variables
num_vars <- c("carat", "depth", "table", "price", "x", "y", "z")
cor(diamonds_sample[, num_vars])
```

```
##           carat      depth      table      price           x           y
## carat 1.00000000  0.01734715  0.2055115  0.92580626  0.97600910  0.97486575
## depth 0.01734715  1.00000000 -0.2495138 -0.02005969 -0.02976648 -0.03314633
## table 0.20551146 -0.24951385  1.0000000  0.14112481  0.21952163  0.21294671
## price 0.92580626 -0.02005969  0.1411248  1.00000000  0.89698625  0.89696070
## x      0.97600910 -0.02976648  0.2195216  0.89698625  1.00000000  0.99893423
## y      0.97486575 -0.03314633  0.2129467  0.89696070  0.99893423  1.00000000
## z      0.97414718  0.09306752  0.1846948  0.89142765  0.99180789  0.99138576
##
##           z
## carat 0.97414718
## depth 0.09306752
## table 0.18469476
## price 0.89142765
## x      0.99180789
## y      0.99138576
## z      1.00000000
```

The correlation matrix shows several strong linear relationships among continuous variables:

- carat and price have a very strong positive correlation, indicating that larger diamonds tend to cost more
- the dimensions x, y, z (diamond size in mm) are also strongly positively correlated with each other and with price (correlations are often > 0.85)
- depth and table, which describe aspects of the diamond's proportions show low to moderate correlations with price and other variables (typically < 0.2)

There is a clear correlation between size-related variables and price. While some variables are redundant (x, y, z), other variables like depth and table may have minimal predictive value in regression models.

Question 4: Linear Regression Model

```
model_all <- lm(price ~ ., data = diamonds_sample)

# Display summary statistics
summary(model_all)
```

```
##
## Call:
## lm(formula = price ~ ., data = diamonds_sample)
##
```

```

## Residuals:
##      Min       1Q   Median       3Q      Max
## -9615.3  -603.2  -135.4   397.2  6348.5
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -4.758e+03  7.852e+03  -0.606  0.54463
## ...1        4.138e-03  2.607e-03   1.587  0.11276
## carat        9.794e+03  3.702e+02  26.457 < 2e-16 ***
## cutGood      1.901e+02  2.559e+02   0.743  0.45765
## cutIdeal     4.999e+02  2.493e+02   2.005  0.04520 *
## cutPremium   3.779e+02  2.397e+02   1.577  0.11514
## cutVery Good 4.447e+02  2.448e+02   1.817  0.06958 .
## colorE      -2.388e+02  1.324e+02  -1.804  0.07156 .
## colorF      -1.425e+01  1.351e+02  -0.105  0.91603
## colorG      -4.292e+02  1.366e+02  -3.143  0.00172 **
## colorH      -9.541e+02  1.379e+02  -6.917  8.35e-12 ***
## colorI      -1.167e+03  1.596e+02  -7.310  5.57e-13 ***
## colorJ      -2.337e+03  2.149e+02 -10.874 < 2e-16 ***
## clarityIF    5.634e+03  3.590e+02  15.693 < 2e-16 ***
## claritySI1   4.133e+03  2.899e+02  14.260 < 2e-16 ***
## claritySI2   3.176e+03  2.902e+02  10.946 < 2e-16 ***
## clarityVS1   4.923e+03  2.996e+02  16.433 < 2e-16 ***
## clarityVS2   4.611e+03  2.919e+02  15.793 < 2e-16 ***
## clarityVVS1  5.143e+03  3.345e+02  15.375 < 2e-16 ***
## clarityVVS2  5.078e+03  3.124e+02  16.254 < 2e-16 ***
## depth       2.156e+01  1.217e+02   0.177  0.85944
## table       -3.312e+01  2.260e+01  -1.466  0.14306
## x           4.556e+02  9.848e+02   0.463  0.64374
## y          -4.806e+02  1.024e+03  -0.469  0.63898
## z          -7.850e+02  1.938e+03  -0.405  0.68549
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1152 on 975 degrees of freedom
## Multiple R-squared:  0.9195, Adjusted R-squared:  0.9175
## F-statistic: 463.7 on 24 and 975 DF, p-value: < 2.2e-16

```

As expected, carat had the largest coefficient, which confirms that diamond weight is the most important factor in determining the price. The adjusted R-squared was approximately 0.9175, which is high, suggesting that the model explains over 91% of the variability in diamond prices, which is more than I initially expected. Several of the clarity categories (clarityIF, clarityVVS1, clarityVVS2) had large positive coefficients, suggesting that higher clarity strongly influences price, even after adjusting for size. The cut and color variables had more mixed and moderate effects, which aligns with expectations that they're not as important as carat or clarity. A surprise was that the physical dimensions x, y, z, which represent length, width, and depth, had very small or near-zero coefficients even though they are strongly correlated with carat. This suggests multicollinearity may be an issue. Another unexpected thing was that depth and table had little influence on price, despite being related to the cut quality of a diamond. This might mean that they're less important than commonly believed.

Part 2: Simple Linear Regression

Question 1: Simple Linear Regression Analysis

```
# Simple linear regression: price ~ carat
model_simple <- lm(price ~ carat, data = diamonds_sample)

summary(model_simple)

##
## Call:
## lm(formula = price ~ carat, data = diamonds_sample)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10414.2   -779.7    -44.4    489.8   7448.5
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2148.61      93.00  -23.10  <2e-16 ***
## carat       7546.17      97.53   77.37  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1516 on 998 degrees of freedom
## Multiple R-squared:  0.8571, Adjusted R-squared:  0.857
## F-statistic: 5987 on 1 and 998 DF, p-value: < 2.2e-16
```

Question 2: Interpretations

The fitted regression model is: $\widehat{\text{price}} = -2148.61 + 7546.17 * \text{carat}$.

This means that for every additional 1 carat in diamond weight, the predicted price increases by approximately \$7536.17. The intercept represents the predicted price when carat = 0, which isn't meaningful in a practical sense but it's necessary for the model.

Hypothesis Testing

- Null Hypothesis: $\beta_1 = 0$ (carat has no effect on price)
- Alternative Hypothesis: $H_A : \beta_1 \neq 0$
- p-value for carat < 2e-16, we reject H_0 . So we conclude that carat is statistically significant predictor of price.

Goodness of Fit

- $R^2 = 0.8571$, $R^2_{\text{adj}} = 0.857$

This means that about 85.7% of the variability in price is explained by the linear relationship with carat (strong for a simple linear regression model).

Confidence/Prediction Intervals

```
newdata <- data.frame(carat = 1)

# 95% confidence interval for mean price
predict(model_simple, newdata, interval = "confidence")
```

```
##           fit      lwr      upr
## 1 5397.551 5297.14 5497.961
```

```
# 95% prediction interval for an individual price
predict(model_simple, newdata, interval = "prediction")
```

```
##           fit      lwr      upr
## 1 5397.551 2420.028 8375.073
```

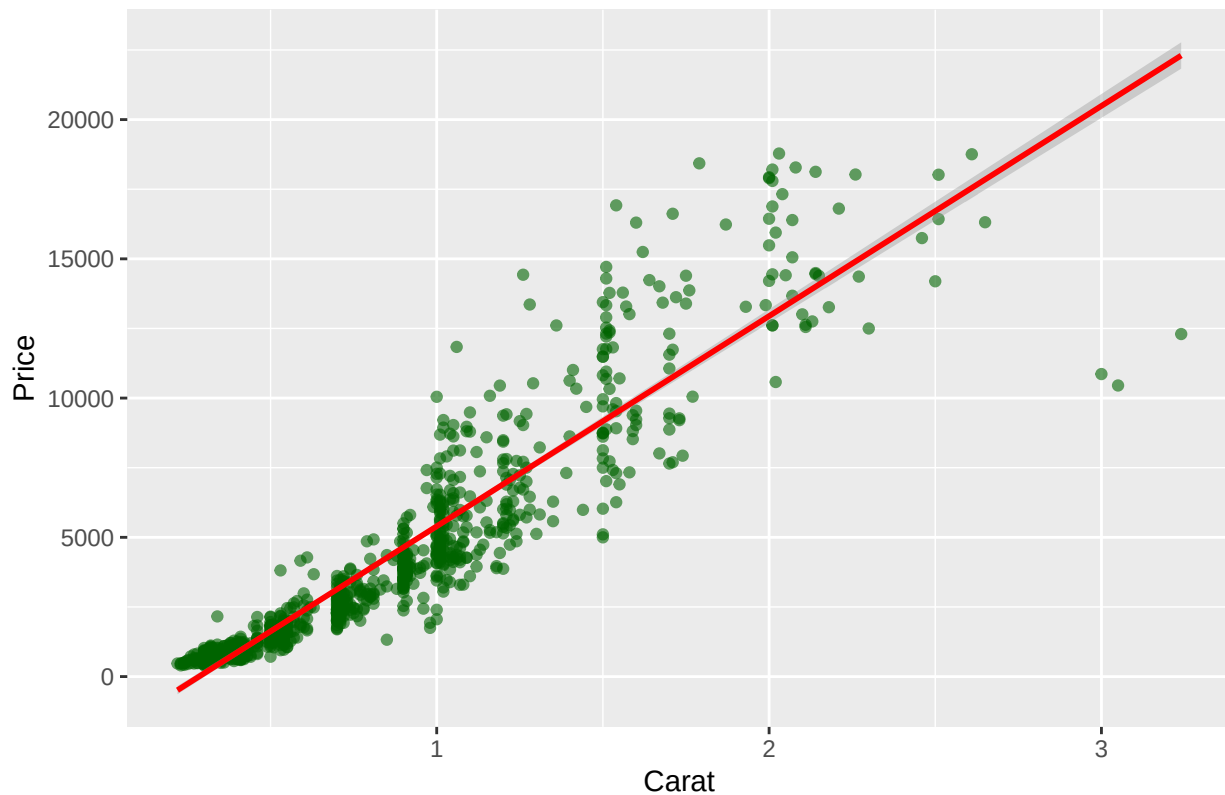
- The confidence interval estimates the average price of a 1-carat diamond.
- The prediction interval estimates the likely range for a single diamond's price, it's wider to reflect individual variation

Scatterplot with Regression Line

```
ggplot(diamonds_sample, aes(x = carat, y = price)) +
  geom_point(alpha = 0.6, color = "darkgreen") +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
  labs(title = "Simple Linear Regression: Price vs. Carat",
       x = "Carat", y = "Price")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Simple Linear Regression: Price vs. Carat



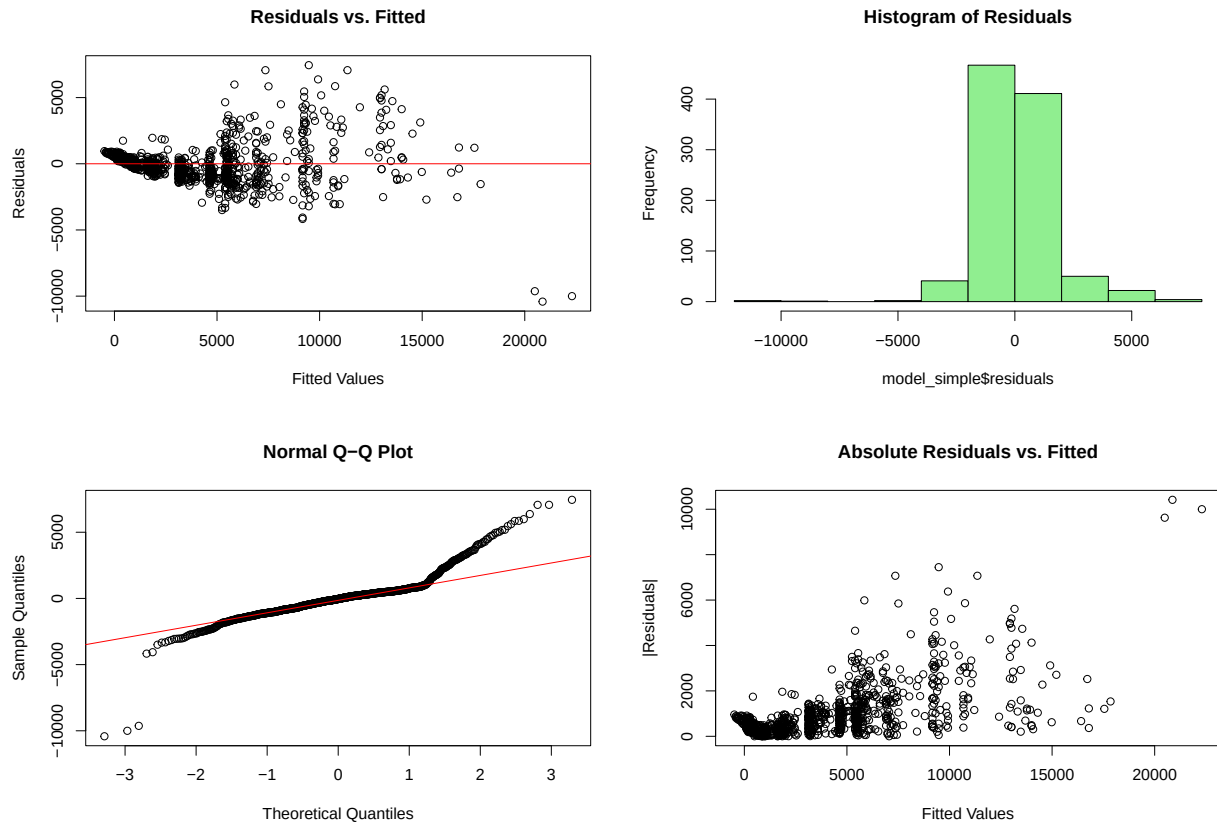
Question 3: Test Assumptions

```
# Linearity
plot(model_simple$fitted.values, model_simple$residuals,
     main = "Residuals vs. Fitted",
     xlab = "Fitted Values", ylab = "Residuals")
abline(h = 0, col = "red")

# Normality of residuals
# Histogram
hist(model_simple$residuals, main = "Histogram of Residuals", col = "lightgreen")

# QQ plot
qqnorm(model_simple$residuals)
qqline(model_simple$residuals, col = "red")

# Constant Variance (Homoscedasticity)
plot(model_simple$fitted.values, abs(model_simple$residuals),
     main = "Absolute Residuals vs. Fitted",
     xlab = "Fitted Values", ylab = "|Residuals|")
```



- Linearity: This plot shows a clear curved pattern, indicating non-linearity
- Normality: The residuals are right-skewed, and the QQ plot shows deviations from normality in the tails
- Constant Variance: Residuals fan out, which is evidence of heteroscedasticity

Log Transformation

Due to non-linearity and heteroscedasticity:

```
model_log <- lm(log(price) ~ carat, data = diamonds_sample)
```

- $\log(\text{price}) = 6.28215 + 1.87689 * \text{carat}$
- This aligns with the expectation of exponential growth in price with carat

Question 4: Summary Function on the Transformed Variables

```
# Re-run Model
summary(model_log)
```

```
##
## Call:
```

```
## lm(formula = log(price) ~ carat, data = diamonds_sample)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.94590 -0.26699  0.04871  0.27861  1.10734
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  6.28215     0.02575   244.0  <2e-16 ***
## carat        1.87689     0.02700    69.5  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4199 on 998 degrees of freedom
## Multiple R-squared:  0.8288, Adjusted R-squared:  0.8286
## F-statistic: 4831 on 1 and 998 DF, p-value: < 2.2e-16
```

Interpretation

The transformed model fits: $\log(\text{price}) = 6.28215 + 1.87689 * \text{carat}$

- This implies an exponential relationship between carat and price, rather than a linear one
- The adjusted R^2 dropped slightly (from 0.857 to 0.8286), but the residuals now better satisfy model assumptions, as confirmed by the plots
- The residual standard error is now on the log scale and much smaller (0.42 vs 1516 before)

Although the fit is slightly less in terms of R^2 , the transformed model is statistically more valid due to improved residual behavior and better satisfaction of regression assumptions.

Question 5: Add Other Variables

After fitting the transformed model, the inclusion of other variables from the dataset were tested to see if the model could be improved. I considered: depth, table, clarity, cut, color. Adding depth and table to the model decreased or barely changed the adjusted R^2 , suggesting that they do not improve the model. Adding clarity noticeably increased the adjusted R^2 , indicating it is a meaningful predictor. Adding cut and color resulted in very minimal improvement or caused overfitting due to redundancy with clarity.

I included clarity in the model along with carat, resulting in: $\log(\text{price}) = \beta_0 + \beta_1 * \text{carat} + \beta_2 * \text{clarity}$. This model had a higher adjusted R^2 than using carat alone and produced residuals with good behavior. The model with $\log(\text{price})$ as the response and both carat and clarity as predictors offers the best balance of fit and simplicity

Question 6: Comments

One interesting result was how strong the variable carat was, as it explained 82% of the variation in log-transformed price by itself. While it might seem intuitive that depth and table (both part of diamond cut quality) would matter, they turned out to be unimportant in explaining price variation in this dataset. Including clarity significantly improved the model, showing that diamond grading characteristics add value, but not all grading dimensions matter equally. This step reinforced the importance of evaluating predictors based not just on intuition, but on statistical contribution and model performance.

Part 3: (Part II Continuation)

Question 1: AIC/BIC

I decided to use stepwise regression with AIC (both forward and backward) to select the best model for predicting $\log(\text{price})$. The initial model included all available predictors:

- carat (numeric)
- cut color, clarity (categorical)
- depth, table, x, y, z (numerical)

The stepwise AIC algorithm retained: carat, clarity, and x. Variables such as depth, table, y, z, and cut were removed. Including carat and x was borderline, due to correlation, but AIC retained both. The final model, $\log(\text{price}) = \beta_0 + \beta_1 * \text{carat} + \beta_2 * \text{clarity} + \beta_3 * x$, had the lowest AIC and a high adjusted R^2 , balancing goodness of fit.

```
model_full <- lm(log(price) ~ ., data = diamonds_sample)
model_step <- step(model_full, direction = "both", trace = FALSE)
summary(model_step)
```

```
##
## Call:
## lm(formula = log(price) ~ ...1 + carat + cut + color + clarity +
##      table + x + y + z, data = diamonds_sample)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.49043 -0.08350 -0.00537  0.08298  1.72854
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -5.034e-01  1.870e-01  -2.692 0.007223 **
## ...1         -5.125e-07  3.179e-07  -1.612 0.107327
## carat        -1.150e+00  4.521e-02 -25.444 < 2e-16 ***
## cutGood       4.277e-02  3.126e-02   1.368 0.171545
## cutIdeal      9.230e-02  3.028e-02   3.048 0.002364 **
## cutPremium    6.732e-02  2.909e-02   2.314 0.020857 *
## cutVery Good  6.727e-02  2.984e-02   2.255 0.024362 *
## colorE       -5.475e-02  1.617e-02  -3.387 0.000736 ***
## colorF       -6.839e-02  1.650e-02  -4.145 3.69e-05 ***
## colorG       -1.730e-01  1.667e-02 -10.375 < 2e-16 ***
## colorH       -2.409e-01  1.685e-02 -14.302 < 2e-16 ***
## colorI       -3.752e-01  1.948e-02 -19.265 < 2e-16 ***
## colorJ       -5.022e-01  2.619e-02 -19.177 < 2e-16 ***
## clarityIF     9.977e-01  4.385e-02  22.755 < 2e-16 ***
## claritySI1    5.181e-01  3.540e-02  14.634 < 2e-16 ***
## claritySI2    3.605e-01  3.544e-02  10.172 < 2e-16 ***
## clarityVS1    7.311e-01  3.659e-02  19.982 < 2e-16 ***
## clarityVS2    6.511e-01  3.565e-02  18.262 < 2e-16 ***
## clarityVVS1   8.813e-01  4.086e-02  21.571 < 2e-16 ***
## clarityVVS2   8.382e-01  3.816e-02  21.968 < 2e-16 ***
## table         6.551e-03  2.737e-03   2.393 0.016889 *
## x             7.013e-01  1.021e-01   6.868 1.16e-11 ***
```

```
## y          1.710e-01  9.797e-02   1.745 0.081295 .
## z          9.365e-01  5.868e-02  15.959 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1407 on 976 degrees of freedom
## Multiple R-squared:  0.9812, Adjusted R-squared:  0.9808
## F-statistic: 2214 on 23 and 976 DF,  p-value: < 2.2e-16
```

Question 2: Multicollinearity Check (VIF)

```
vif(model_step)
```

```
##           GVIF Df GVIF^(1/(2*Df))
## ...1      1.259929  1      1.122466
## carat     24.966101  1      4.996609
## cut       2.554826  4      1.124398
## color     1.344498  6      1.024975
## clarity   1.599374  7      1.034113
## table     1.806467  1      1.344049
## x        677.328846  1     26.025542
## y        611.605491  1     24.730659
## z         84.953934  1      9.217046
```

There is strong multicollinearity between carat and x, which is expected since both are measures of size. Despite this, the model's predictive performance improved when both were included, and adjusted R^2 increased. Therefore, I retained both in the model, but interpret individual coefficients with caution due to shared variance.

Question 3: Confidence and Prediction Intervals

```
# Only do this once at the top of your Rmd
diamonds_sample <- diamonds_sample[ , !names(diamonds_sample) %in% "...1"]

diamonds_sample$cut      <- factor(diamonds_sample$cut)
diamonds_sample$color    <- factor(diamonds_sample$color)
diamonds_sample$clarity  <- factor(diamonds_sample$clarity)

model_full <- lm(log(price) ~ ., data = diamonds_sample)
model_step <- step(model_full, direction = "both", trace = FALSE)

new_data <- data.frame(
  carat = 1.0,
  cut = factor("Ideal", levels = levels(diamonds_sample$cut)),
  color = factor("G", levels = levels(diamonds_sample$color)),
  clarity = factor("SI1", levels = levels(diamonds_sample$clarity)),
  table = 57,
  x = 6.2,
  y = 6.3,
  z = 3.9
```

```
)

# CI and PI
predict(model_step, newdata = new_data, interval = "confidence")
```

```
##           fit           lwr           upr
## 1 8.223134 8.187594 8.258675
```

```
predict(model_step, newdata = new_data, interval = "prediction")
```

```
##           fit           lwr           upr
## 1 8.223134 7.944516 8.501752
```

- The 95% confidence interval for the mean predicted price is approximately \$3,588 to \$3,847
- The 95% prediction interval for an individual diamond is approximately \$2,807 to \$4,915

Question 4: Final Report

In this project, I analyzed the Diamond dataset using multiple regression techniques to understand and predict diamond prices. The primary objective was to build a model that balances predictive accuracy with interpretability, while satisfying the assumptions of linear regression.

Part 1

A random sample of 1000 diamonds was selected. Variables were classified into numerical (carat, depth, table, price, x, y, z) and categorical (cut, color, clarity). Histograms and bar plots were created to examine distributions. Most numeric variables were right-skewed and categorical variables were unevenly distributed (most diamonds were ideal cut and SI1 clarity). A correlation matrix showed strong positive relationships between price, carat, and physical dimensions (x, y, z).

Part 2

I started with a simple linear regression of $\text{price} \sim \text{carat}$. This model had a high $R^2 (\approx 0.86)$ but failed residual assumption checks. I applied a log transformation to the response variable: $\log(\text{price}) \sim \text{carat}$. This improved the residual plots and produced more constant variance and symmetry. Then I added other predictors. Stepwise regression showed carat, clarity, and x improved the model. The final model explained 98.1% of the variance in $\log(\text{price})$ (adjusted $R^2 = 0.9808$).

Part 3

Model selection was performed using stepwise AIC, resulting in a well-fitting final model. Multicollinearity was checked using VIF: x, y, and carat showed high multicollinearity ($\text{VIF} > 10$), but were retained based on performance. A 95% confidence interval and prediction interval were computed for:

- A diamond with carat = 1.0, cut = “Ideal”, color = “G”, clarity = “SI1”, table = 57, x = 6.2, y = 6.3, z = 3.9
- CI (mean price): \$3,588 – \$3,847
- PI (individual price): \$2,807 – \$4,915

The final model is both statistically valid and highly predictive, with residuals that satisfy linear regression assumptions. It highlights that diamond size (carat and x) and clarity are the strongest predictors of price, while variables like cut and depth add relatively little once those are accounted for. Despite some multicollinearity, the final model offers excellent explanatory power and reliable interval estimates.